MUSTAFA MAHMOUD

AI-NATIVE ANALYTICS ENGINEER | AI DEVELOPER | REVENUE OPERATIONS ANALYST

Building systems with agentic AI

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Open to remote and relocation opportunities

PROFESSIONAL SUMMARY

AI-native analytics engineer and AI developer who builds systems with agentic AI, with a revenue operations background and 4+ years turning commercial questions into reliable data products. Directs AI coding agents to build and run analytics at speed, and verifies every output with TRUST, a framework created and applied by me: trace the sources, record the rules, underwrite the numbers, ship, track. Built an operations intelligence and planning platform, a revenue forecasting engine that backtests six methods to pick the best, and an unattended daily data pipeline for a multi-region sales organisation, covering SQL extraction, validation, attribution, reconciliation and automated reporting. Gave the organisation daily WhatsApp visibility it never had, by reverse-engineering the BI platform's own data endpoints when no API was available. Works with teams across Egypt, China and the Gulf.

CORE SKILLS

Revenue Operations & Analytics: Revenue forecasting and statistical modelling (time-series methods, exponential smoothing, rolling-origin backtesting, WAPE/MAE model selection, bias tracking), funnel and cohort analysis, revenue-driver decomposition, attribution rules, KPI design, target pacing, root-cause analysis
Data & SQL: SQL (CTEs, deduplication, windowed aggregation, reconciliation), openpyxl, data validation, control totals, Microsoft Excel (pivot tables, advanced formulas, Power Query, Power Pivot), DAX data modelling (the Power BI and Power Pivot engine)
Data Engineering & Integration: Python, JavaScript, unattended daily pipelines, reverse-engineered BI endpoints (backend access without an API), Playwright browser automation, Git and GitHub, automated testing, Cloudflare (Pages, Workers, D1)
AI & Automation: TRUST verification framework (my own), AI agent workflows (Claude Code, Codex), agentic loops (inspect, implement, test, refine), prompt and skill design, multi-agent runs, n8n, Gemini-assisted workflows
Stakeholders & Presentation: Stakeholder management across Egypt, China and the Gulf, executive-ready data presentation, data visualisation and dashboard design, business intelligence (BI) reporting, data video with Remotion, ECharts dashboards

PROFESSIONAL EXPERIENCE

Sales Strategy, Analytics Systems and AI Automation Specialist | 51Talk | Cairo, Egypt **Jul 2026 - Present**

Operations Intelligence & Planning Platform (flagship)

- Designed and built a bilingual (English/Chinese) analytics platform with 12 connected modules for the Egypt and Jordan adult business, turning daily CSV, ZIP, order, target and service exports into governed snapshots. It connects revenue, acquisition, conversion, referrals, team performance and WhatsApp in one daily workflow, so stakeholders can investigate gaps and plan without requesting a new report.
- Adopted across leadership: a weekly average of 8 stakeholders, 20 opens, 130 page views and about 27 hours of active use, with owner activity excluded.
- Defined the attribution and metric rules: event-in-period versus registration cohorts, orders linked by student ID, approved channel logic, Beijing-to-Saudi date conversion and double-count prevention. Every rate is recomputed from filtered numerators and denominators, never averaged.
- Built reconciliation controls (Total to Region to Team to CC), deduplication and unmatched-activity QA that keeps unmatched records visible instead of silently dropping them.
- Added target pacing views for leadership.
- Run releases through a gated workflow: automated JavaScript and Python tests, real-browser validation, bilingual checks and documentation sync, with pull-request-only promotion from staging to production.

Revenue Forecasting Engine

- Built a Python forecasting engine that makes six methods compete, from naive and moving-average baselines to exponential smoothing, a damped-trend model and a driver-based model (paid orders times average order value). It picks the winner on evidence: rolling-origin backtests ranked by WAPE, then MAE, then simplicity, with bias tracking and historical error ranges.
- Landed within 0.97% of actual monthly revenue in its September backtest.
- Extended it with a lead-aware in-month model using lead volume, conversion timing, weekday patterns and revenue already booked, built through an agentic loop and challenged against transparent baselines before release.

Automated Daily Data Pipeline

- Brought 16 reporting views into one automated daily workflow, with an unattended pipeline that collects email attachments and BI exports, validates 22 source inputs, builds the snapshot, publishes the platform and notifies stakeholder groups without duplicate sends.
- Built a validator that flags malformed source files (for example, a line with 24 fields instead of 133) and automatically alerts the data engineering team with the exact file, line and counts, and notifies the source owner.
- **Biggest unlock: daily WhatsApp data with no API.** I had no API access to the BI platform, and nobody could see WhatsApp data day to day. Where others stopped at "not doable", I proposed reverse-engineering the platform's own data endpoints, an idea none of the AI tools had suggested; the agents then audited how the interface loads its data and rebuilt the call. A headless Chrome session reuses an authorised login, takes a fresh security token and runs the SQL query from inside the browser, with nobody logging in. WhatsApp data now arrives every morning. Each extraction takes about 8 seconds and is accepted only if schema, date coverage, row counts and control totals agree.
- Built the pipeline so that a retry can never send the same alert twice.
- Took over the daily sales data report in about a week. It now runs unattended every morning as the official daily dashboard, checked against the reports it replaced before cut-over.

AI-Agent Workflows

- Created TRUST (Trace, Record, Underwrite, Ship, Track), a framework for verifying AI-assisted analytics: source profiling, written metric specs the agents follow, control-total and reference-dataset diffs before release, then freshness checks and failure alerts in production. It supported the cut-over of the daily dashboard.
- Ran an agentic loop (inspect, implement, test, refine) to rebuild a production data validator that halted daily runs on already-reviewed defects. Set the boundaries: scoped inputs, real-export replay as the finish line, and a non-publishing mode.
- Added a defect register that tracks each reviewed issue per record. The deterministic Python run now warns on known defects, emails the data engineering team automatically to investigate newly missing records, and still blocks on integrity failures, with the acceptance rules decided by me.
- Wrote operating rules for AI agents: role split, handoff protocols and cost controls for multi-agent runs.

Quality and Data Analyst | Creo Global | Remote, Dubai UAE **Aug 2025 - May 2026**

- Built a measurement framework that quantified contribution by failure type and exposed recurring platform defects; WhatsApp reachability rose from 66% to 80% within three weeks.
- Created a daily KPI and weekly close-out system that turned campaign deviations into specific issues for operational review.
- Worked across a remote, multi-timezone team to connect performance analysis with process and compliance decisions.

SELECTED PROJECTS

- **AI lead qualification and routing:** n8n workflow connecting lead capture, eligibility checks, programme routing, tailored email, Google Sheets logging and Gemini-assisted recommendations for sales review. Cut a 15-20 minute manual process to seconds per lead.
- **Performance root-cause analysis:** Connected multiple data sources, identified the three main failure points and isolated the behavioural drivers behind a sharp metric decline. The metric recovered six percentage points within one week and the analysis became team training material.
- **Performance segmentation:** Four-week rolling segmentation model for coaching. Top-tier representation rose from 24% to 29% and from 19% to 24% across two channels within three weeks.

EARLIER EXPERIENCE

Quality Evaluator | Concentrix **Oct 2023 - Jan 2026**

Turned recurring performance problems into structured analyses and decision tools for operational leaders. Built automated Excel systems with multi-source validation that replaced manual tracking.

EDUCATION AND CERTIFICATIONS

Bachelor's Degree in English Literature and Linguistics | International College for Languages and Translation | 2021
Intermediate SQL | DataCamp
EF SET English Certificate | C1 Advanced, 68/100 | Nov 2025
Languages | Arabic native | English C1 Advanced